Chu Liyun

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github.com/MKurisu004



Research Interests

Machine learning systems; efficient LLM inference; AI infrastructure.

Education

Tsinghua University Beijing, China
B.E. in Information and Computational Science Sep. 2023 – Jun. 2027 (Expected)

Research Experience

Undergraduate Researcher 2025 – Present
OmniLab, Tsinghua University, Beijing, China
Advisor: Prof. Ju Ren

- Proposed *Octopus*, an SSD-based KVCache offloading system optimized for parallel I/O across multiple SSDs. The paper is accepted by ACM SIGOPS ATC' 26.
- Reduced I/O time by 2.41× and improved effective bandwidth utilization by 2.72×.

Intern Jan. 2026 – Jan. 2027 (Expected)
AI Infra Team, General AI Research Center, Ant Research Institute

- Optimize FP8 reinforcement learning rollout in Ant's agentic RL framework, Asystem.
- Currently working on RSI (recursive self-improvement) evaluation infrastructure.

Publications

- Tuowei Wang*, **Liyun Chu***, Ruwen Fan, and Ju Ren. *Octopus: Unlocking SSD Parallelism for Sparse KVCache Retrieval.* ACM SIGOPS ATC 2026. Paper

Honors and Awards

- MLSys 2026 Competition Track: AWS Trainium2/3 MoE Kernel Challenge, Top 3 (2026).
- Outstanding Scholarship for Science and Technology Innovation, 2025.
- Second-Class Award, Basic Interdisciplinary Challenge Competition, Zhili College, 2025.
- Freshman Scholarship, Tsinghua University, 2023–2025.

* Equal contribution.